🌦️ MONSOON — FULL MIND MAP

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1. 'Monsoon' Word Origin — Arabic

- 'Monsoon' word **Arabic word 'mausim' (matlab 'season')** se aaya hai — seasonal wind-reversal ke roop mein South/South-East Asia mein rainfall ka reason maana jaata hai. *[Q1]*

2. Himalaya's Climatic Role (Assertion-Reason)

- India ek monsoonal country hai (Assertion sahi) kyunki **Himalaya ek effective climatic divide ka kaam karta hai** — cold northern winds se shield deta hai, aur monsoon winds ko trap karke unhein subcontinent mein hi moisture chhodne majboor karta hai (Reason sahi explanation hai). *[Q2]*

3. India's Tropical Monsoon Climate — Latitude Facts

- India ki climate **tropical monsoon type hai**, **8°4' se 37°6' N latitude** ke beech hai — lekin **exactly tropical latitudes (30°N-30°S) ke beech NAHI** hai (Assertion sahi, Reason galat). *[Q4]*

4. Monsoon Arrival — Kerala First

- India mein monsoon **Kerala mein sabse pehle** aata hai — SW monsoon se **India ki total rainfall ka 75%** milti hai, June-September active rehta hai. *[Q5]*

5. Monsoon Wind Direction (SW to NE)

- Summer monsoon **South-West se North-East** direction mein behta hai — winter mein retreat karte waqt **North-East se South-West**. Monsoon ki origin **South-West winds** se hoti hai — high temperature/low pressure Indian subcontinent par Indian Ocean se air khींchta hai. *[Q6, Q7, Q8]*

6. Sea Salinity Rankings (Red Sea Highest)

- Salinity ranking: **Red Sea (41‰, sabse zyada, landlocked hone ki wajah se) > Mediterranean Sea (high evaporation) > North Sea (North Atlantic Drift se) > Baltic Sea (sabse kam, river-water influx ki wajah se)**. Normal open ocean salinity **33-37‰** hoti hai. *[Q9]*

7. Climate Diagram Regions (NE India)

- Heavy summer rainfall + moderate temperature wala climate diagram **North-East India** ko represent karta hai. *[Q10]*

8. Driest Place — Leh

- Mumbai, Delhi, Leh, Bengaluru mein **Leh sabse shushk (driest) jagah** hai — least rainfall receive karta hai. *[Q11]*

9. January Isotherm — 18°C Zonal Divide

- India ko tropical aur subtropical zones mein baantne wala **January isotherm 18°C** hai. Given map ke shaded area mein July ka mean temperature **25.0°C-27.5°C** hota hai (annual mean temperature ka map North India ko South se garm dikhata hai). *[Q3, Q12, Q13]*

10. Daily Temperature Range — Rajasthan Desert Highest

- India mein **Rajasthan ke desert areas mein sabse zyada daily temperature range** hota hai — din mein garam, raat mein thanda (desert climate ki main feature). *[Q14]*

11. North-East Monsoon — Tamil Nadu's Rainy Season

- Tamil Nadu ka main rainy season **November-December (North-East Monsoon, October-December)** hai — Andhra Pradesh, Rayalaseema, Tamil Nadu mein is period mein rainfall hoti hai. Isi monsoon mein **26 December 2004 ko Tamil Nadu mein Tsunami aayi thi**. *[Q15]*

12. Monsoon Origin Theories (Thermal & Dynamic)

- Monsoon ki seasonal displacement **land aur sea ke differential temperature** ki wajah se hoti hai. Do main concepts: **(1) Thermal concept — differential heating effect ka result**; **(2) Dynamic concept (Flohn ne diya) — planetary winds aur pressure belts ke seasonal migration ka result**. *[Q16]*

13. East-West Rainfall Gradient & Duration Gradient

- Northern plains mein annual rainfall **east se west ki taraf GHATTI hai** (winds ki humidity kam hoti jaati hai) — aur **South India mein monsoon duration North se zyada lamba** hota hai (sea ke kareeb hone ki wajah se) — dono statements sahi hain. *[Q17]*

14. Indian Ocean Dipole (IOD) & El Niño/La Niña

- IOD ('Indian Niño') — **western aur eastern Indian Ocean ke beech sea-surface temperature ka irregular oscillation** hai (tropical Eastern Pacific Ocean ke SAATH NAHI, ye galat statement hai) — **IOD, El Niño ke monsoon-impact ko influence kar sakta hai (ye sahi hai)**. *[Q18]*

- El Niño (Peru coast ki warm ocean current) **Indian Monsoon ko kamzor karta hai — harr saal NAHI, balki 2-7 saal ke irregular interval mein aata hai**. La Niña iska opposite hai, jo monsoon ko strengthen karta hai. *[Q19]*

⚠️ EXAM TRAP — IOD — western vs eastern Indian Ocean ke beech hota hai, Pacific Ocean ke saath NAHI; El Niño har saal nahi aata

15. Jet Stream — Tibetan Plateau Split

- Jet stream Himalaya ke north mein Tibetan highlands ke parallel Asia continent ko cross karti hai — Tibet ki hills se **do branches mein split ho jaati hai (north aur south)** — lekin **northern branch India ke winter weather mein important role NAHI nibhati (southern branch nibhati hai)**. *[Q20]*

16. Altitude vs Latitude — Agra-Darjeeling, Amritsar-Shimla

- Agra aur Darjeeling same latitude par hain lekin January temperature Agra mein 16°C, Darjeeling mein sirf 4°C hai — **altitude ke saath temperature ghatta hai (normal lapse rate: 1°C per 165m)** isi ka reason hai. Isi tarah **Amritsar-Shimla ki climate difference bhi altitude ke difference se hai** (Amritsar ~756 feet, Shimla ~7866 feet). *[Q21, Q22]*

17. Climate Regions of India (Humid SE, Sub-Humid)

- Chhattisgarh mein **'Humid South-East' climate** hai (West Bengal, Chotanagpur, Odisha plain, southern Chhattisgarh, NE Andhra Pradesh mein bhi milti hai) — **Sub-humid transitional — central Gangetic plains, Sub-humid littoral — Coromandel, Sub-humid continental — upper Gangetic plains**. *[Q23]*

18. Arabian Sea Branch — Chhattisgarh Basin Unaffected

- Monsoon ki Arabian Sea branch, Western Ghats se obstruct hoke **2 branches mein split hoti hai** — ek Aravalli ke parallel (scanty rainfall), dusri Vindhya se obstruct hoke Gujarat-MP-Western Chhattisgarh mein rainfall karti hai. **Chhattisgarh Basin isse zyada affected NAHI hai**. *[Q24]*

19. Humid Climate Cities (Kochi, Tejpur)

- Ahmedabad, Kochi, Ludhiana, Tejpur mein se **Kochi aur Tejpur zyada humid hain** (heavy rainfall ki wajah se — 2022 mein Kochi 3415.8mm, Tejpur 1775.7mm, jabki Ahmedabad sirf 564mm, Ludhiana 766.9mm). *[Q25]*

20. Monsoon Withdrawal (Retreat) Indicators

- Monsoon ki decline/withdrawal 3 cheezon se indicate hoti hai: **clear sky, Bay of Bengal mein low pressure ban na, land par temperature ka thoda badhna** — sab teen sahi hain. *[Q26]*

21. Westerlies & Western Disturbances

- Westerlies **30°-60° north AND south latitudes** ke beech behti hain (30°N se 60°S ka statement galat hai) — **winter mein western cyclonic disturbances (Mediterranean se origin) westerly jet stream se India laayi jaati hain, North-West India mein winter rain karti hain**. *[Q27]*

22. ITCZ — Equatorial Low Pressure

- Intertropical Convergence Zone (ITCZ) **equator par** hoti hai — NE aur SE trade winds yahin milti hain, sun ki apparent motion ke saath north-south shift karti hai. *[Q28]*

23. Rajasthan Dust Storms — Causes

- Rajasthan mein May-June ke dust storms ka main reason **convectional currents ka origin (kuch jagahon par)** hai — 'Loo' garam hawa se low pressure banta hai, jisse air rapidly rise karke dust storms laati hai. *[Q29]*

24. Rajasthan's Low Rainfall — Aravalli's Role

- Rajasthan mein bahut kam rainfall hoti hai kyunki **winds koi barrier cross nahi karti jo unhein cool hone ke liye zaroori uplift de sake** — Aravalli range SW monsoon ki direction ke parallel align hai, isliye partial obstacle bhi effective nahi hai, aur western Rajasthan rain-shadow area ban jaata hai. *[Q30]*

25. Hindu Calendar Seasons — Chronological Order

- Hindu calendar ke 6 seasons chronological order mein: **Spring (March-April) → Summer (May-June) → Rainy (July-August) → Autumn (September-October) → Pre-winter (November-December) → Winter (January-February)**. *[Q31]*

26. MONEX — Indo-Soviet Monsoon Expedition

- India-USSR ne **1973 mein joint 'Monsoon Expedition' MONEX-1973** ki thi. 1967 mein WMO-ICSU ne **Global Atmospheric Research Program (GARP)** organize kiya tha, jiske **Global Weather Experiment (1 Dec 1978)** ke sub-program ke roop mein **Monsoon Experiment (MONEX-1979)** tha — dono alag events hain (Expedition vs Experiment ka confusion trap). *[Q32]*

27. Koppen Classification — Cwg for Ganga Plains

- Koppen ne India ke Great Plains (upper Gangetic plains) ke liye **'Cwg' (dry-winter humid subtropical, 'Monsoon type with dry winters')** use kiya — summer temperature ~40°C se winter mein ~27°C tak girta hai. **North-East India (North Bengal samet) aur North Bihar dono ka bhi Koppen classification 'Cwg' hai**. *[Q33, Q34, Q35]*